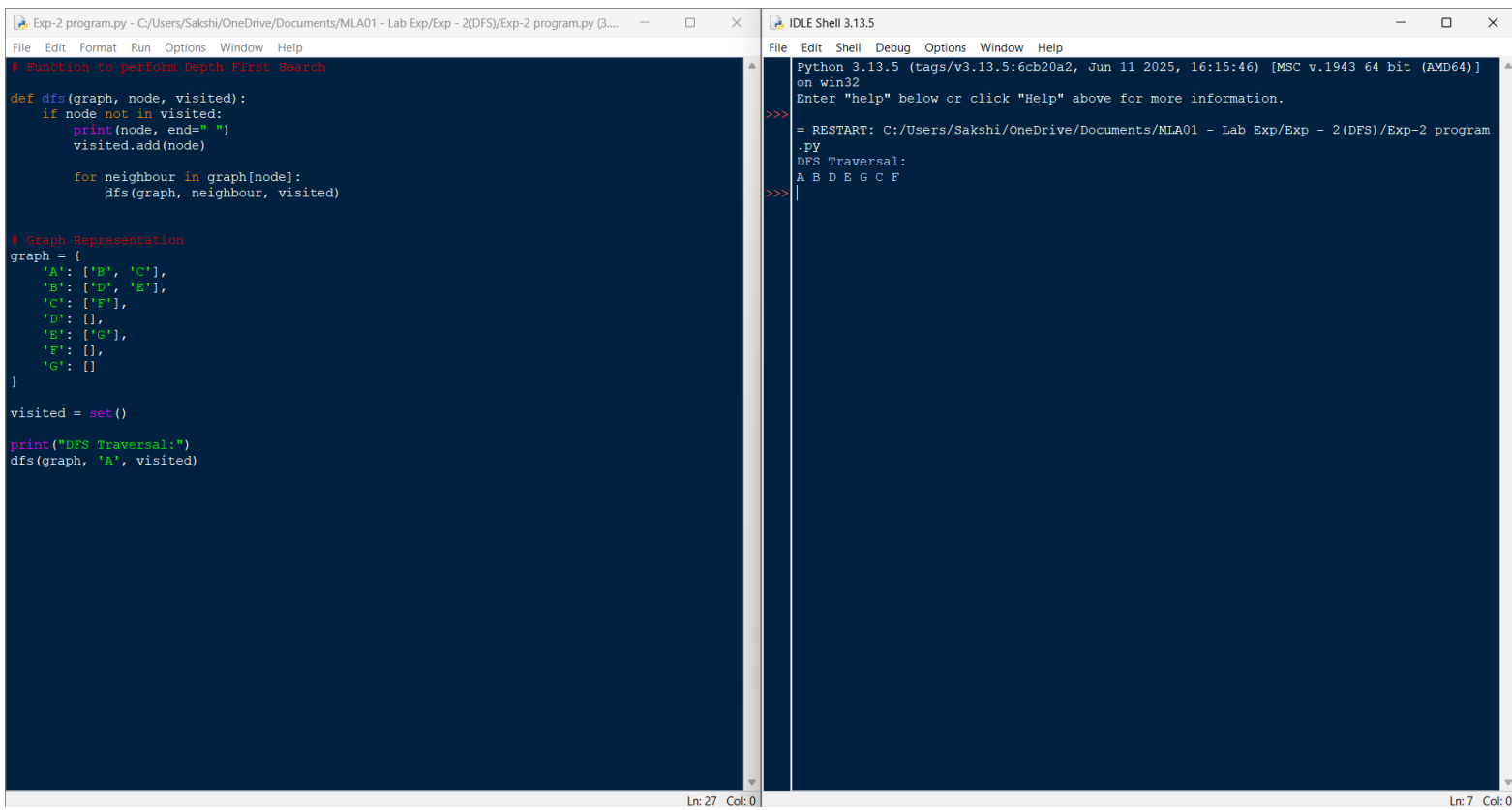


Exp:2 DFS

Program:

```
def dfs(graph, node, visited):  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
        for neighbour in graph[node]:  
            dfs(graph, neighbour, visited)  
  
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['G'],  
    'F': [],  
    'G': []  
}  
  
visited = set()  
  
print("DFS Traversal:")  
dfs(graph, 'A', visited)
```

Output:



The image shows a screenshot of an IDE with two windows. The left window, titled 'Exp-2 program.py', contains a Python script for a Depth First Search (DFS) algorithm. The script defines a recursive function 'dfs' that traverses a graph, prints the nodes in the order they are visited, and adds them to a 'visited' set. The graph is represented as a dictionary with nodes 'A' through 'G' and their neighbors. The right window, titled 'IDLE Shell 3.13.5', shows the output of the program: 'DFS Traversal: A B D E G C F'.

```
# Function to perform Depth First Search
def dfs(graph, node, visited):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for neighbour in graph[node]:
            dfs(graph, neighbour, visited)

# Graph Representation
graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': ['G'],
    'F': [],
    'G': []
}

visited = set()

print("DFS Traversal:")
dfs(graph, 'A', visited)
```

```
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp - 2(DFS)/Exp-2 program
.PY
DFS Traversal:
A B D E G C F
>>>
```